Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030478

NAME: ADAMA SAMARTH

Query 1: - CREATE DATABASE

```
2520030478-Adama Samarth
CREATE DATABASE IF NOT EXISTS joins_db;
USE joins_db;
```

Output: -

	#	Time	Action	Message	Duration / Fetch
✔	1	13:55:57	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected	0.031 sec
✔	2	13:56:01	USE joins_db	0 row(s) affected	0.000 sec

Query 2: -CREATE TABLES

```
CREATE TABLE class (
    id INT,
    name VARCHAR(30)
);
CREATE TABLE class_info (
    id INT,
    address VARCHAR(30)
);
```

Output: -

	#	Time	Action	Message	Duration / Fetch
✔	3	13:58:04	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected	0.125 sec
✔	4	13:58:07	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected	0.047 sec

Query 3: -INSERT VALUES

```sql
25200:30478-Adama Samarth
INSERT  INTO class VALUES
(1,'Adama Samarth'),
(2,'Rohit Verma'),
(4,'Vivek Sharma');

INSERT INTO class_info VALUES
(1,'HYDERABAD'),
(2,'DELHI'),
(3,'CHENNAI');
```

Output: -

	#	Time	Action	Message	Duration / Fetch
✔	7	14:01:03	INSERT INTO class VALUES (1,'Adama Samarth'), (2,'Rohit Verma'), (4,'Vivek Sharma')	3 row(s) affected	0.031 sec
✔	8	14:01:08	INSERT INTO class_info VALUES (1,'HYDERABAD'), (2,'DELHI'), (3,'CHENNAI')	3 row(s) affected	0.062 sec

Query 4: - CROSS JOIN

```sql
25200390478-Adama Samarth
SELECT *
FROM class
CROSS JOIN class_info;
```

Output: -

✔	17	14:04:24	SELECT * FROM class CROSS JOIN class_info	9 row(s) returned	0.000 sec / 0.000 sec

Result Grid | Filter Rows: | Export:

id	name	id	address
1	Adama Samarth	1	HYDERABAD
2	Rohit Verma	1	HYDERABAD
4	Vivek Sharma	1	HYDERABAD
1	Adama Samarth	2	DELHI
2	Rohit Verma	2	DELHI
4	Vivek Sharma	2	DELHI
1	Adama Samarth	3	CHENNAI
2	Rohit Verma	3	CHENNAI
4	Vivek Sharma	3	CHENNAI

Result 1 ×

Query 5: - INNER JOIN

```sql
25200030478-Adama Samarth
SELECT c.id, c.name, ci.address
FROM class c
INNER JOIN class_info ci ON c.id = ci.id;
```

Output: -

✔	18	14:08:29	SELECT * FROM class INNER JOIN class_info O...	2 row(s) returned	0.000 sec / 0.000 sec

Result Grid Filter Rows:

	id	name	id	address
▶	1	Adama Samarth	1	HYDERABAD
	2	Aditya Rao	2	DELHI

Query 6: -INNER JOIN

Output: -

✔	19	14:10:36	SELECT class.name, class_info.address FRO...	2 row(s) returned	0.000 sec / 0.000 sec

Result Grid Filter Rows:

	name	address
▶	Adama Samarth	HYDERABAD
	Aditya Rao	DELHI

Query 7: - NATURAL JOIN

Output: -

✔	20	14:12:02	SELECT * FROM class NATURAL JOIN class_info	2 row(s) returned	0.000 sec / 0.000 sec

Result Grid Filter Rows:

	id	name	address
▶	1	Adama Samarth	HYDERABAD
	2	Aditya Rao	DELHI

Query 8: - LEFT OUTER JOIN

Output: -

```
21  14:14:30  INSERT INTO class VALUES (5,'Nihitha')                          1 row(s) affected                          0.031 sec
22  14:14:36  INSERT INTO class_info VALUES (7,'BENGALURU'),(8,'PUNE')        2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0  0.046 sec
```

Result Grid | Filter Rows: | Export:

	id	name	id	address
▶	1	Adama Samarth	1	HYDERABAD
	2	Aditya Rao	2	DELHI
	3	Neha Sharma	NULL	NULL
	5	Nihitha	NULL	NULL

Query 9: - LEFT JOIN

Output: -

```
23  14:16:23  SELECT * FROM class LEFT OUTER JOIN class_info ON class.id=class_info.id  4 row(s) returned  0.000 sec / 0.000 sec
```

Result Grid | Filter Rows: | Export:

	id	name	id	address
▶	1	Adama Samarth	1	HYDERABAD
	2	Aditya Rao	2	DELHI
	3	Neha Sharma	NULL	NULL
	5	Nihitha	NULL	NULL

Query 10: - LEFT JOIN

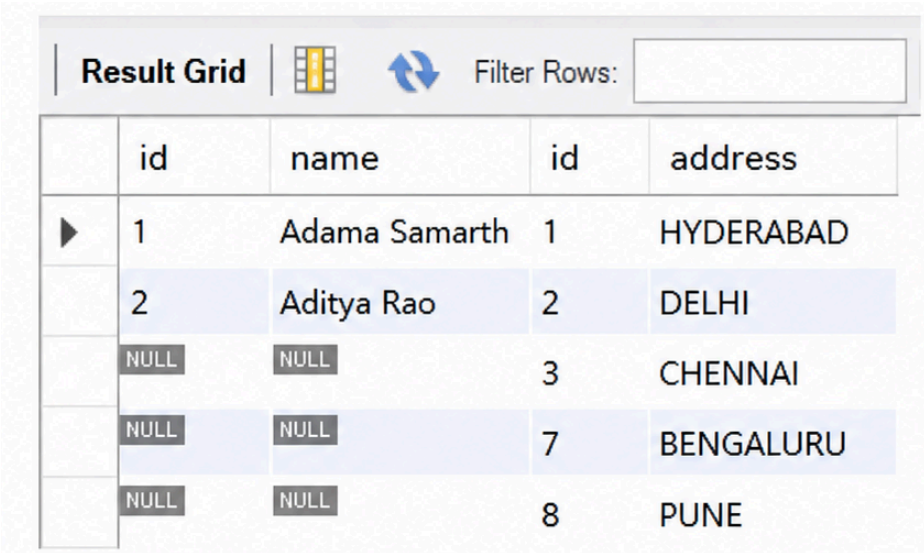
Output: -

```
24  14:18:42  SELECT * FROM class_info LEFT JOIN class ON class_info.id=class.id  4 row(s) returned  0.000 sec / 0.000 sec
```

Result Grid | Filter Rows: | Export:

	id	address	id	name
▶	1	HYDERABAD	1	Adama Samarth
	2	DELHI	2	Aditya Rao
	7	BENGALURU	NULL	NULL
	8	PUNE	NULL	NULL

Query-11



id	name	id	address
1	Adama Samarth	1	HYDERABAD
2	Aditya Rao	2	DELHI
NULL	NULL	3	CHENNAI
NULL	NULL	7	BENGALURU
NULL	NULL	8	PUNE

Query-12: Union

```
2520030478-Adama Samarth
CREATE TABLE first_table(
    roll_no INT,
    name VARCHAR(30)
);
CREATE TABLE second_table(
    roll_no INT,
    name VARCHAR(30)
);
INSERT INTO first_table VALUES
    (101,'Rahul'),
    (102,'Vikram');
INSERT INTO second_table VALUES
    (102,'Vikram'),
    (103,'Arjun');
```

✓	28	14:25:46	CREATE TABLE first_table(roll_no INT, name VARC...	0 row(s) affected	0.047 sec
✓	29	14:25:50	CREATE TABLE second_table(roll_no INT, name VA...	0 row(s) affected	0.031 sec
✓	30	14:25:54	INSERT INTO first_table VALUES (101,'Rahul'), (10...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.046 sec
✓	31	14:25:58	INSERT INTO second_table VALUES (102,'Vikram'),...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.047 sec

```
SELECT * FROM first_table
UNION
SELECT * FROM second_table;
```

✓	32	14:27:30	SELECT * FROM first_table UNION SELECT * FR...	3 row(s) returned	0.031 sec / 0.000 sec

Result Grid | Filter Rows:

	roll_no	name
▶	101	Rahul
	102	Vikram
	103	Arjun

Query 13: - UNION ALL

```
2520030478-Adama Samarth
SELECT * FROM first_table
UNION ALL
SELECT * FROM second_table;
```

✓	33	14:28:55	SELECT * FROM first_table UNION ALL SELECT ...	4 row(s) returned	0.031 sec / 0.000 sec

Result Grid | Filter Rows:

	roll_no	name
▶	101	Rahul
	102	Vikram
	102	Vikram
	103	Arjun

Query 14: -INTERSECT (MYSQL DO NOT SUPPORT INTERSECT DIRECTLY)

```
SELECT first_table.*
FROM first_table
INNER JOIN second_table
ON first_table.id = second_table.id;
```

Output: -

34 14:32:12 SELECT first_table.* FROM first_table INNER JOI... 1 row(s) returned

	id	name
▶	2	Manasvi

Query 15: -MINUS (MYSQL DO NOT SUPPORT MINUS DIRECTLY)

```
SELECT first_table.*
FROM first_table
LEFT JOIN second_table
ON first_table.id = second_table.id
WHERE second_table.id IS NULL;
```

Output: -

35 14:34:55 SELECT first_table.* FROM first_table LEFT JOIN... 1 row(s) returned

ROLL NO: 2520030478 - Adama Samarth

Query 16: -Generate a report showing all students and address availability status.

```
-- 2520030478 - Adama Samarth
SELECT c.id,
       c.name,
       CASE
           WHEN ci.address IS NULL
           THEN 'Address Missing'
           ELSE 'Address Available'
       END AS Status
FROM class c
LEFT JOIN class_info ci
ON c.id=ci.id;
```

Output: -

37 14:37:51 SELECT c.id, c.name, CASE WH... 3 row(s) returned 0.000 s

	id	name	Status
▶	1	Akshay	Address Available
	2	Manasvi	Address Available
	3	Ramya	Address Missing

Query 17: -FULL OUTER JOIN (MYSQL DO NOT DIRECTLY SUPPORT FULL OUTER JOIN)

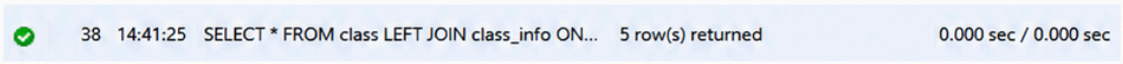
ROLL NO: 2520030478 - Adama Samarth

```
-- 2520030478 - Adama Samarth
SELECT *
FROM class
LEFT JOIN class_info
ON class.id = class_info.id
WHERE class_info.id IS NULL

UNION

SELECT *
FROM class
RIGHT JOIN class_info
ON class.id = class_info.id
WHERE class.id IS NULL;
```

Output: -



	id	name	id	address
▶	4	Amulya	NULL	NULL
	5	Nihitha	NULL	NULL
	NULL	NULL	3	CHENNAI
	NULL	NULL	7	BENGALURU
	NULL	NULL	8	PUNE